Examiner only

8. *(a)* Some elements in the *p*-block form compounds where the *p*-block atom does not have eight electrons in its outer shell.

(i) Explain why nitrogen can only form a chloride with eight outer shell electrons, but phosphorus can form a chloride with a different number of outer shell electrons. [2]

You should give the chemical formulae of relevant compounds in your answer.

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(ii) Explain why aluminium forms compounds that are electron deficient. Show how one of these compounds can act to gain a full outer shell. [2]

You should give the dot and cross diagrams of the electron deficient species.

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(b) Below is some information about the oxides of two Group 4 elements.

Carbon	There are two common oxides of carbon. Carbon dioxide is an acidic oxide and carbon monoxide can be used as a reducing agent. Both of these are gases with very low boiling temperatures.
Lead	There are two common oxides of lead. Lead(II) oxide is an amphoteric oxide and lead(IV) oxide can be used as an oxidising agent. Both of these are solids that exhibit a large degree of ionic character.

Explain the differences between the oxides of carbon and lead, giving chemical equations to illustrate their acid/base and redox properties. [6 QER]

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